

Grasping Data: Mapping Out HCI Methods for Children and Young People's Interactions with their Personal Data

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Abstract

Children and young people today are uniquely datafied, tracked, and digitally monitored like no generation before them. Children are increasingly vulnerable to data collection through seemingly benign toys and devices equipped with voice recognition, geolocation, sensors, and cameras, but typically lack awareness or control over these data exchanges, as consent is usually provided by adult caregivers (who may also lack data literacy). Although there have been advancements in legislation and design guidelines, there is a significant need for interdisciplinary approaches to directly empower children in understanding, valuing and benefiting from their personal data, as well as learning to manage it. We ask, how can the CHI community support children to be informed and included in design for and with their personal data? How do we support children to participate in their personal data – and do they want to? This workshop aims to gather designers, researchers and practitioners working on projects relating to children's personal data, and to map out the current practices and methods at play across the CHI community on this critical topic.

CCS Concepts

• **Human-centered computing** > **Interaction design** > **Interaction design process and methods** > **Participatory design**;

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Children, Data, Personal Data, Participatory Methods

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1 Motivation

This workshop seeks to determine the lay of the land in the CHI community when it comes to designing for and with children's personal data. Despite a growing public awareness of technology's accelerating capacity to capture, process and exploit personal data, this has not been matched by efforts to promote understanding of the myriad ways in which our information is harvested [2]. This is of greater concern when it comes to children, as the abstract nature of data and lack of clear consensus on data education, means that children are often not aware of the risks and consequences of personal data processing and how it can be used or misappropriated. Children are increasingly exposed to risks through their use of apparently innocuous toys and devices that can generate data from everyday interactions through voice recognition, geolocation software, sensors, and cameras [8]. This enables third parties to 'know' a child's birthday, heart rate or whereabouts, to monitor tooth brushing and attention levels, or capture personal conversations and play routines. Children lack agency in these transactions, as consent for their data is either acquired through adult caregivers (who themselves may lack data literacy) or not at all. Whilst some

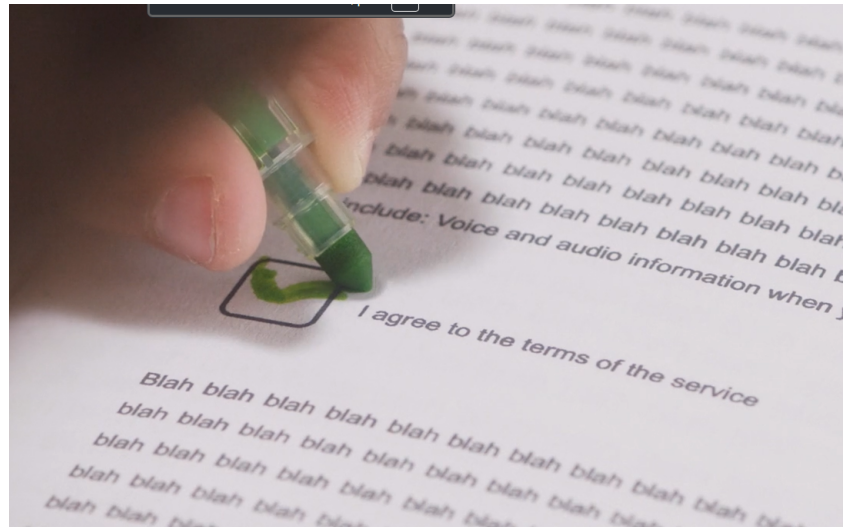


Figure 1: Screenshot from 5Right’s Twisted Toys Provocation - Available at <https://twistedtoys.5rightsfoundation.com/>

progress has been made in tackling this issue through legislation, regulation and even design guidelines [3], there is a dearth of interdisciplinary work exploring how to empower children directly. Children are often considered too young to understand these complexities, however their worlds are increasingly encroached upon by personal data tensions, particularly with the recent proliferation of artificial intelligence.

Aligned with the General Data Protection Regulation [5], we define personal data as information that is related to an identified or identifiable person in terms of its potential for revealing things about you. Fuelled by concerns around the uncontrolled use and misuse of personal data by third parties, international and national policy (e.g. [5]) recognises the imperative to promote data literacy from the earliest age. Yet, initiatives such as these lack clear pathways for empowering children to understand, value and benefit from their own personal data and support their personal data literacy. In the context of children aged 9 to 16, personal data literacy may include understanding what data is, how it can be generated and interpreted, how to manage it, how to employ tactics to minimize the data given and its implications for self and others [10]. Other work, reviewed by [11], echoes this interest in the over 8s and tends to focus on data that children provide voluntarily, such as their name, rather than other sources of data traces. Research on younger children (aged 3-8yrs) and personal data is very limited, with a noticeable absence of children’s voices. Despite the emergence of pedagogical processes and tools for children’s data literacy [13, 14], there remains a knowledge gap in how to make activities personal, meaningful, and engaging for young children.

HCI is uniquely placed to address some of these pressing issues. As a community we engage with many varied and creative participatory methods and approaches to understanding personal data contexts, for example with adults, [6, 12, 14] and some with children [4, 7, 9]. Concurrently, designers and researchers on child rights are using speculative, provocative and playful methods. 5Rights

Foundation [1] have probed how children’s agency over their personal data is obscured in the everyday with a range of provocations called ‘Twisted Toys’ (Fig. 1) where they speculate on “the World of Twisted Toys™ - a wonderland of excitement, experiences and exploitation. We pride ourselves on making toys that are addictive, risky and put you completely under our control” [1]. One key provocation here is “My First Terms and Conditions” which is described as “essential reading for any child. Lose yourself (and your rights) in a few thousand pages of legally impenetrable small print. It’s long, incomprehensible, unfair, and deliberately obscure – almost like it was designed not to be read...”. Thus, we see interesting attempts at probing and unsettling the status quo on children’s personal data coming from various HCI-related fields through engaging, provocative methods.

Indeed, some of the organizers of this workshop have begun to explore these issues through innovative methods – such as data physicalisation [15] - in a new UKRI funded interdisciplinary grant. The “Grasping Data” project addresses the societal need for personal agency in an increasingly datafied world, recognizing that, as society’s youngest members, children’s futures are most impacted by the imbalance between those who give and profit from personal data. Here we are exploring how we tackle theoretical questions about how we make abstract concepts concrete, and advancing methods for physicalizing real-time dynamic data so that it can be manipulated and investigated by children. We look to be developing versatile physicalisation toolkits for schools, new data-driven exhibit prototypes for public knowledge institutions such as museums and science centres, and physicalisation activities for extending informal learning experiences. We note that our approach is one of many possible routes to understanding children’s personal data with them, and this workshop provides an opportunity for mutual learning on best practice and future directions.

1.1 Questions the workshop will pose

- How do we support children to participate in / with / through their personal data?
- What are colleagues' existing work and interests around personal data and CYP?
- What methods are being used across CHI to understand this?
- What steps can we as a community take to ensure children are kept safe and yet have agency over their data?
- How are children being involved (or not) in explorations of their personal data?
- How is children's agency over their personal data obscured in the everyday?
- What works, with which age groups, and in which design contexts?

1.2 Workshop Goals

- 1) To understand where and how children are being included in research and design in the CHI community which concerns itself with their personal data.
- 2) To create an explorative visual mapping of the methods and mindsets currently used by the CHI community around children and personal data and in which context.
- 3) To converge on a set of principles or a manifesto for the design community to follow on best practice in such contexts.

2 Organisers

Dr Cara Wilson is an Assistant Professor in Children and Technology at the Centre for Research in Digital Education and at the Edinburgh Futures Institute at the University of Edinburgh. With a background in participatory design with children and young people, she explores how to better support inclusion and diversity in the development of technologies which pervade children's lives, for example, working with neurodiverse children in primary school contexts. Through creative, explorative and participatory design methods, her research supports the strengths, voice and agency of individuals who are often overlooked in the process of design. Cara is a co-lead on the UKRI Grasping Data grant exploring methods to support children to understand and use their personal data, and the Bridging Responsible AI Divides grant. She has been lead- and co-organiser for several successful workshops across CHI and other ACM venues, such as IDC and PDC.

Professor Andrew Manches is Full Professor of Children and Technology at the University of Edinburgh, where he leads the Children and Technology group as co-director of the Digital Education Research Centre. Since moving to academia from a career as a classroom teacher, Andrew has led multiple funded projects examining the role of body-based interaction in the way children think and learn and the implications for new forms of technology. He is the Project Lead for 'Grasping Data', a £1.2million UKRI-funded project (2025-2026) looking at the potential to help children understand and value their personal data. Andrew works closely alongside practitioners and designers and has spun out two companies to scale impact from design work. In 2021, Andrew was awarded the Tam Dalyell prize for excellence in engaging the Public with science to recognize his ongoing efforts to bridge his work with everyday educational practice and design.

Ayca Atabey is a Research Associate and PhD candidate at the University of Edinburgh. She is an affiliate at the Centre for Data, Culture & Society of the Edinburgh Futures Institute. She conducts interdisciplinary research at the intersection of Data Protection Law and Value-Sensitive Design in Human-Computer Interaction, focusing on fairness in design and regulation of AI-driven EdTech. She received a PhD Enrichment Award from the Alan Turing Institute, where she worked on Child-Centred AI with a focus on ethical use of personal data and design of data-intensive technologies. She also has worked at the Digital Futures Commission at 5Rights Foundation focusing on beneficial uses of personal data in education, EdTech evaluation frameworks and child rights and privacy-respecting design of online services. She has previously co-organized CCAI workshops at CHI (2023, 2024).

Dr Uta Hinrichs is a Reader in Data Visualization at the University of Edinburgh where she co-leads VisHub – interdisciplinary research group focusing on teaching, collaboration, and outreach around data visualization hosted by the Institute for Design Informatics and the School of Informatics. Uta's research focuses on the design and engagement with data in the widest sense through interactive technologies and data visualization and related methods. Heavily drawing from fields such as Design, Literary Studies, and Information Sciences, Uta's research has been published at leading venues in HCI, visualization, and the humanities. She has co-led a large range of data physicalisation workshops in different settings, exploring a variety of processes, activities, and materials to engage both adults and children with both personal data as well as those linked to political, social or economic phenomena.

Professor Steve Brewster is Professor of Human-Computer Interaction in the School of Computing Science at the University of Glasgow and co-chaired the CHI conference in 2019. He leads the Multimodal Interaction Group there, focusing on multimodal HCI, or using multiple sensory modalities and control mechanisms (particularly audio, haptics and gesture) to create a rich, natural interaction between human and computer. He holds a current ERC Advanced Grant investigating (AR/VR) and a previous EPSRC Advanced Research Fellowship (haptics/multimodal access to data for blind/visually-impaired learners). His work has a strong experimental focus, applying perceptual research to practical situations, currently turning to children's personal data through novel physicalisation methods.

Dr Ge Wang is a Postdoctoral Fellow at Stanford University in the Human-Centred Artificial Intelligence group. Her research focuses on Child-Computer AI interaction, and investigates the algorithmic impact on families and children, exploring the potential for designing more age-appropriate and autonomy-supportive AI for families. She led the first CCAI workshop at CHI 2023, and has experience co-organising several workshops, seminars and forums, and served as a volunteer in several previous conferences and workshops, for example in her previous role at the University of Oxford.

Dr Bernd Ploderer is an Associate Professor in Human-Computer Interaction at Queensland University of Technology (QUT) in Brisbane, Australia. Working primarily in health and wellbeing domains, Dr Ploderer studies, designs and evaluates interactive technologies to help people make sense of data about

Table 1: Example Workshop Schedule

Time	Activity
9.30am – 9.45am	Welcome
9.45am – 11.15am	Activity 1
11.15am – 11.30am	Break
11.30am – 1pm	Activity 2
1pm – 3pm	Lunch
3pm – 4.30pm	Activity 3
4.30pm – 5pm	Wrap Up and Next Steps
5pm	Dinner TBC

themselves and the world around them. He is an Associate Investigator at the Australian Research Council Centre of Excellence for the Digital Child, where he works with children and parents to co-design technologies for better sleep and active play.

3 Workshop Logistics

3.1 Plans to publish workshop proceedings

Papers will be published on the workshop website and on arXiv, with a view to expanding these for a special edition journal.

3.2 Hybrid delivery

To promote equitable participation, we will host a hybrid workshop, taking into account visa restrictions and sustainability. The hybrid format enhances global participation by reducing the relevance of travel costs and visa concerns for remote attendees and supports sustainability by reducing travel. We aim to enhance the experience for both in-person and remote participants – something which our organizing committee has rich experience in delivering effectively. Remote participants will engage with those on-site through various platforms, including our website, Slack channel, Miro boards, and Zoom or Microsoft Teams. We have found in our previous experience running CHI and other ACM workshops that hybrid formats support many more people to attend and this will be very helpful in our mapping exercise to gather as many views and methods as possible from participants globally.

3.3 Accessibility

Accessibility is a core concern for our organizing committee, and we will strive to create an inclusive, welcoming environment for people of all abilities. This includes physical accessibility for those in-person (e.g. ensuring the venue is wheelchair accessible, ensuring that it is suitable for people with sensory sensitivities, ensuring subtitling if requested) and also digital accessibility for all (e.g. strongly advising participants to adhere to the SIGCHI's Guide to an Accessible Submission). We will inquire about any specific needs in advance so as to ensure participants' individual needs can be supported on the day.

3.4 Asynchronous engagement

Synchronous engagement will be through video-conference and online collaboration tools. For video-conference, participants will join via Zoom or Microsoft Teams. We will create a Miro board

and will share links with participants to join the boards during the workshop. Should asynchronous engagement be needed, we will disseminate a protocol that will guide participants to go through the activities in their own time. We will include screenshots of the Miro boards and examples of use. In addition, we will set up a Slack server for asynchronous communication within participants and organizers.

3.5 Workshop Schedule and Activities

3.5.1 Activity One: How are we currently involving children in designs leveraging their personal data? This will work towards addressing Goal 1- *to understand where and how children are being included in research and design in the CHI community which concerns itself with their personal data* (see 1 example workshop schedule).

In advance of the workshop, we will group participants into thematic groups of 4-5 based on their submissions. Two weeks prior to the workshop, we will ask participants to read each other's position papers and propose questions. On the day, we will begin with a playful icebreaker within each group and a concise topic-focused introduction from each group member. Each activity will have an assigned notetaker. We will begin by asking each participant to share what they submitted in their position paper briefly – beginning with *the question they would most like to ask a child about their thoughts on their personal data*.

We will then work – either with post-its and pens (then photographed and uploaded for online access), or directly in Miro – to discuss and map out the commonalities and differences in each other's work. These will be guided by some of the core questions of this proposal, such as what agency could children have over their personal data? We will focus on the methods and approaches which participants are engaging with and the outcomes these are yielding. At the end of this session, each group will have created a thematic mapping based on their own work which we will discuss together as a larger group.

3.5.2 Activity Two: What are the big issues for design here and how do we map them out? This activity will work towards addressing Goal 2 - *to create an explorative visual mapping of the methods and mindsets currently used by the CHI community around children and personal data and in which contexts*.

Based on the discussions and sharing from Activity 1, we will use the second session to tease out the 'big issues' in this space with colleagues globally – some framings we will use to probe

concern child rights, consent, and data literacy. We will build on the smaller group maps from Activity 1 to create a 'Methods and Mindsets Map'. This will be conducted as a whole group discussion and ideation session, which we have experience in facilitating with a hybrid approach. We will strive to produce something useful for the community through this workshop, hence we will employ a graphic designer /artist on the day to help visualize this mapping in an accessible and digestible way, which will be subsequently polished and shared widely.

3.5.3 Activity Three: What can we do better? This will work towards addressing Goal 3 - *to converge on a set of principles or a manifesto for the design community to follow on best practice of research in such contexts.*

To begin, we will return to our small groups and create five key messages we would want to share with the CHI community on working with children and their personal data. These will be probed by questions such as: What have we found is working, and what have we found that is not? How can we as individual researchers do this better? And, what does design more widely need to do better? Then we will reconvene as a whole group and refine these down to a set of key messages agreed on by all. We believe that these key principles, combined with the visual mapping, will be the foundations of a vibrant community of interest in HCI and children's personal data.

3.6 Post workshop plans

As we are asking participants to submit a 'mini case study' on work they are doing, we see these as the springboard to a special issue journal publication in, for example, the International Journal of Child-Computer Interaction or International Journal of Human-Computer Studies – both of which the organizing committee has had success at publishing special issues from CHI and ACM workshops in the past (such as Pushing the Boundaries of Participatory Design with Children). We are also very keen to build a community of interest on this important topic and plan to continue lively debates using our Slack channel, subsequently using this as a platform to seek collaborative funding opportunities.

3.7 Expected size of attendance

We aim for 15 participants in person and a further 15 online as a minimum. We have had success with previous attendance numbers at CHI workshops (e.g. 2024's Child-Centred AI workshop garnered over 30 participants) through hybrid formats and feel confident that we can garner similar attendance this year through our varied and diverse networks as well as an increasingly topical theme.

4 Call for participation

Despite a growing public awareness of technology's accelerating capacity to capture, process and exploit personal data, this has not been matched by efforts to promote understanding of the myriad ways in which our information is harvested. Children and young people in particular are increasingly exposed to risks through their use of apparently innocuous toys and devices that can generate data from everyday interactions. What methods are being used across CHI to understand this? What steps can we as a community

take to ensure children are kept safe and yet have agency over their data?

The goals of this workshop are:

- 1) to create an explorative visual mapping of the methods and mindsets currently used by the CHI community around children and personal data and in which contexts.
- 2) to understand where and how children are being included in research and design which concerns itself with their personal data.
- 3) to converge on a set of principles or a manifesto for the design community to follow on best practice in such contexts.

This full-day workshop will be delivered in hybrid format to enable remote participation. Potential participants are asked to submit a 4-page (excluding references, in CHI submission format) position paper to the lead author via email, addressing the following as a mini case study:

- An overview of the context currently being explored (age of children, context of fieldwork e.g. schools/homes)
- Methodologies employed
- Some key examples / vignettes / images / data snippets
- Reflections on key challenges and opportunities in research around children and personal data
- One questions I would like to ask participants working on this critical topic
- One question I would like to ask children about the topic of their personal data

Participants will be selected on the basis of their position paper suitability and on ensuring diversity of attendance from participants across fields, career stages, and experience. Accepted papers will be published on the workshop website, hosted at CHI 2025 Workshop "Grasping Data: Mapping Out HCI Methods for Children and Young People's Interactions with their Personal Data" | Digital Education. It is a requirement that at least one author of each accepted submission must attend the workshop and that all participants must register for both the workshop and for at least one day of the conference. **Submissions are due by Friday 7th February 2025 midnight (AoE).**

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